

SciOdyssey.

A research layer over your agent harness.

PRESERVE THE RESEARCH WORLD · RESET THE RESEARCHER

ROBUST

SUPERVISABLE

BROAD SEARCH

SAME EVIDENCE
FRESH JUDGMENT
A WIDER SEARCH

Research
Runs
Longer.

01 / PROBLEM

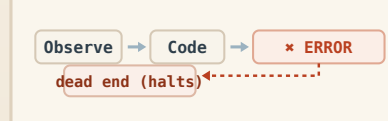
Why current agents fall short.

Sustained autonomous research breaks in three different ways.

01

Closed-world fragility

Fixed workflows shatter on unexpected runtime errors.



Unhandled runtime faults halt unassisted runs

02

Trajectory momentum

A single context carries past momentum into local basins.



22 runs trapped tweaking parameters in same basin

03

Context inflation cost

Reconstructing context costs far more than the edit itself.



10 files read to change 1 line; burns token budget

DESIGN RESPONSE

Keep the world. Reset the trajectory.

02 / SYSTEM

One persistent world. Fresh scientific judgment.

Give the research world and the researcher different lifetimes.

DUAL-LIFETIME LOOP

Persistent world · ephemeral trajectory · audited write-back

Fresh Scientist

Owns scientific judgment inside one trajectory: hypotheses, code, experiments, interpretation, and pivots.

META Review

Audits claims against artifacts, scopes conclusions, and decides what is durable enough to survive.

SELECTIVE PERSISTENCE

Only audited empirical findings, code state, and bounded priors write back to the shared world.

PERSISTENT RESEARCH WORLD

What survives across trajectories

A · EVIDENCE

Knowledge + Ledger
Metrics, failures, reports, and experiment receipts in **ledger.jsonl**.

B · PRIORS

Brief + Hypotheses
Current understanding and exploration leads in **brief.md**, open to revision.

C · STATE

Code + State
Retained implementation, serialized model weights, checks, and git provenance.

CONTEXT RESET

The next Scientist reads verified experience from disk, but does not inherit the previous reasoning trace.

RUNTIME SUBSTRATE

WORKTREE ISOLATION

Independent git worktrees isolate branches and preserve reproducible artifacts.

SELF-HEALING

Traceback → patch → rerun keeps autonomous research moving after runtime faults.

PARALLEL SYNTHESIS

Completed branches can be compared post-hoc; useful ideas survive even when a branch loses.

The next Scientist inherits **verified experience** — not prior reasoning momentum.

Decoupled Memory · Audited Persistence · Fresh Scientific Judgment

03 / EVALUATION

Five empirical claims. Measured evidence.

Standardized evaluation on KuaiRand-Pure benchmark · Fixed official evaluator · Complete telemetry accounting

01 MEASURED RESULT

The model got better.

0.6016000

Official FM baseline

→

0.6059363

SciOdyssey submission

+0.0043363 Primary improvement

GAUC 0.6728421

(+0.0054)

nDCG@5 0.5390304

(+0.0033)

LogLoss 0.38102

(-0.0032)

Feature Engine: 46 categorical cross-interactions + log1p transforms

Architecture: 8-seed Factorization Machine ensemble (NumPy / CPU)

Monotonic gain from baseline (0.6016) → feature engineering (0.6041) → seed ensembling (0.6059); **100% CPU inference** without GPU clusters.

02 SUSTAINED SEARCH

The search stayed productive.

0.60163

C1: Valid Base

0.60409

C2: Features

0.60443

C3: Var. Red.

0.60594

C4: 8-Seed FM

SciOdyssey	0.6059363	Climbed across 4 cycles (7 Full evals)
Codex Goal	0.6044533	Peaked at C8; stalled & dropped in 22 runs

4 autonomous cycles

7 Full evaluations

+0.00148 over Codex

7 Full public-validation evaluations retained across 4 cycles. Trajectory climbed via **structural feature expansion** rather than single lucky edits.

03 SELF-HEALING & GATEKEEPING

The agent recovered from failure.

TIER 1 · RUNTIME SELF-HEALING	CRASH AUTO-RECOVERY
FAILURE ACTION	Cycle 1 numpy.float32 serialization crash Scientist parsed traceback, auto-patched scalar helper
OUTCOME	0 human bugfixes · Unattended rerun succeeded

TIER 2 · META GATEKEEPING	SPURIOUS STATE REJECTION
FALSE CLAIM ACTION	Scientist claimed BPR gain on medium proxy META audited system/data.py, caught UNK leak
OUTCOME	Promotion blocked · Zero contaminated state

Empirical receipts preserved: runtime self-healing telemetry and independent code-level proxy audits.

04 ARCHITECTURE ABLATION

Meta-scientist beats direct agents.

01 · DIRECT Antigravity	02 · SUBAGENTS + Subagents	03 · META-SCIENTIST Persistent World
0.60458	→ 0.60472	→ 0.60520
Single 2h run	Delegated swarm	+ fresh Scientist

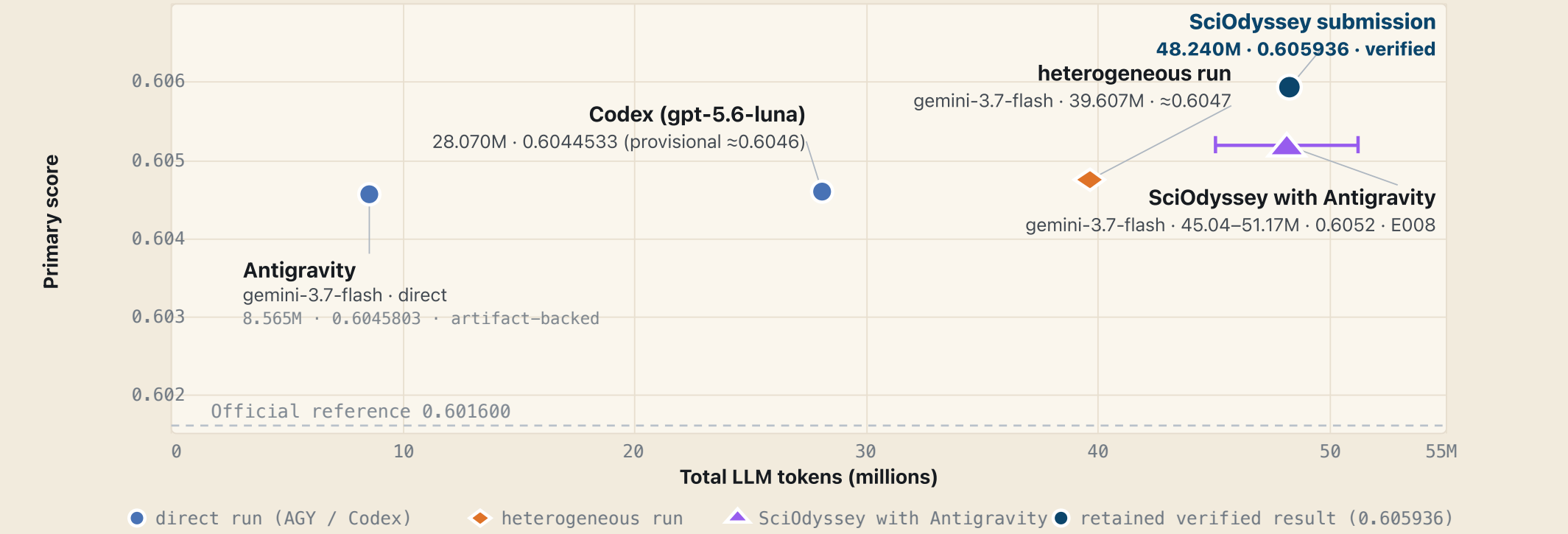
+0.00062 organization gain (Gemini 3.7 Flash)

Keep Gemini 3.7 Flash fixed and change only research organization: **direct agent (0.6046) → subagents (0.6047) → Meta-Scientist (0.6052)**.

05 RESOURCE ACCOUNTING

We are not tokenmaxxing. Score vs token investment.

~\$10 cost · 1 × M2 · 0 GPU-hr · 1h 55m
KuaiRand-Pure · Public validation · Inclusive input+output tokens



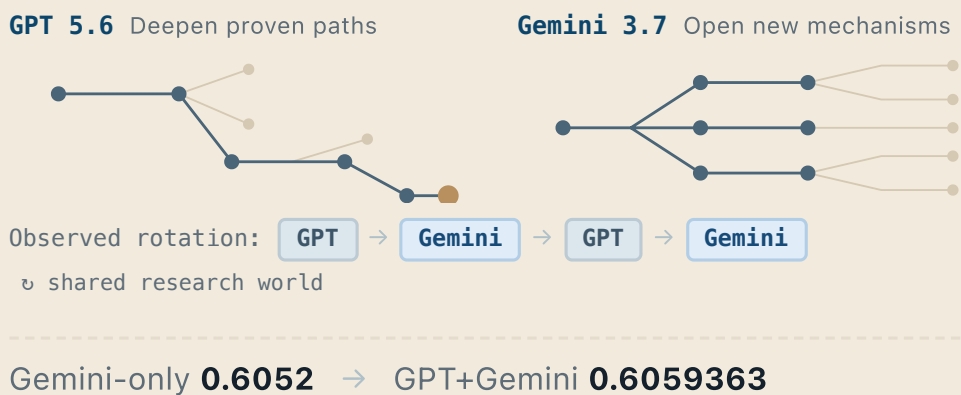
All results **verified** on KuaiRand-Pure official validation split with **complete reproducible artifacts**.

04 / FINDINGS

What the run revealed.

01 · MODEL DIVERSITY

Model diversity can help break plateaus.



Gemini-only 0.6052 → GPT+Gemini 0.6059363

02 · PARALLEL SYNTHESIS

A losing branch can still improve the winner.



Selection keeps useful knowledge, not just the top score: Branch C contributed pairwise ranking and rank normalization.

Branch B+C Ideas → Synthesis 0.6055536

05 / EXPERIENCE

Start your journey.

task.md

What to solve

Formal task contract: research objectives, dataset inputs, compute constraints, and official evaluation protocol.

- Dataset paths & split integrity
- Official sandbox evaluator

PERSONAL.md

How to work

User research preferences, model guidance, exploration style, tool permissions, and supervisory checkpoints.

- Model rotation (GPT / Gemini)
- Supervision & budget thresholds

SUPERVISION TIERS

Tier 1 Unattended Overnight Runs · Tier 2 Checkpoint Review (META) · Tier 3 Steer Working Priors

1. Define task.md → 2. Single-step test → 3. Autonomous search → 4. Parallel explore → 5. Inspect evidence

DECLARATIVE PROTOCOL

Human defines boundary and preferences (task.md + PERSONAL.md); SciOdyssey autonomously iterates, heals, and crystallizes verified evidence.

Autonomous by default.

Supervisable by design.

Coding agent harness · Execution studio

Works with: Codex · Claude Code · Antigravity · OpenCode · Trae

> SKILL.md "Use the research-agent skill. Explore in parallel; let me review what to keep."

\$ research-agent step # one cycle · human reviews

META → Scientist → Evidence → Your turn ✓

\$ research-agent run --max-cycles 4 # autonomous · fresh scientist each round

C1 → C2 → C3 → C4 · Candidate Retained ✓

\$ research-agent parallel --branches 3 # isolated worktrees · reviewer synthesizes

Branch A·B·C → Reviewer Synthesis → Explicit Promotion ✓

\$ research-agent gui # dashboard · telemetry & checkpoints

Telemetry · Loss Curves · Verified Checkpoints ✓

